

8.5 Similarity

The matrix for a linear operator $T : V \rightarrow V$ depends on the basis selected for V . One of the fundamental problems of linear algebra is to choose a basis for V that makes the matrix for T as simple as possible—a diagonal or a triangular matrix, for example. In this section we study this problem.

Simple Matrices for Linear Operators

Standard bases do not necessarily produce the simplest matrices. Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose matrix relative to the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$ is

$$[T]_B = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \quad (1)$$

Compare this with $[T]_{B'}$ for the same operator T but relative to the basis $B' = \{\mathbf{u}'_1, \mathbf{u}'_2\}$, where

$$\mathbf{u}'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}'_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (2)$$

Since

$$T(\mathbf{u}'_1) = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\mathbf{u}'_1 \quad \text{and} \quad T(\mathbf{u}'_2) = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\mathbf{u}'_2$$

it follows that $[T(\mathbf{u}'_1)]_{B'} = [2, 0]^T$ and $[T(\mathbf{u}'_2)]_{B'} = [0, 3]^T$, so

$$[T]_{B'} = [T(\mathbf{u}'_1)_{B'} \mid T(\mathbf{u}'_2)_{B'}] = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

This diagonal matrix is simpler than $[T]_B$ and conveys immediately that T scales $\mathbf{u}'_1$ by a factor of 2 and $\mathbf{u}'_2$ by a factor of 3—information not evident from $[T]_B$.

The strategy for finding the simplest possible matrix for $T : V \rightarrow V$ is to compute $[T]_B$ for any convenient basis, then change the basis to simplify. Before pursuing this, we revisit transition matrices.

A New View of Transition Matrices

Recall from Section 4.7: if $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ and $B' = \{\mathbf{u}'_1, \dots, \mathbf{u}'_n\}$ are bases for V , the transition matrices are

$$P_{B \rightarrow B'} = [\mathbf{u}_1]_{B'} \mid [\mathbf{u}_2]_{B'} \mid \cdots \mid [\mathbf{u}_n]_{B'} \quad (3)$$

$$P_{B' \rightarrow B} = [\mathbf{u}'_1]_B \mid [\mathbf{u}'_2]_B \mid \cdots \mid [\mathbf{u}'_n]_B \quad (4)$$

where $P_{B \rightarrow B'}$ and $P_{B' \rightarrow B}$ are inverses of each other. Also, for any $\mathbf{v} \in V$:

$$P_{B \rightarrow B'}[\mathbf{v}]_B = [\mathbf{v}]_{B'} \quad (5)$$

$$P_{B' \rightarrow B}[\mathbf{v}]_{B'} = [\mathbf{v}]_B \quad (6)$$

The following theorem shows that these transition matrices can be viewed as matrices for the identity operator.

Theorem 8.5.1

If B and B' are bases for a finite-dimensional vector space V , and if $I : V \rightarrow V$ is the identity operator on V , then

$$P_{B \rightarrow B'} = [I]_{B', B} \quad \text{and} \quad P_{B' \rightarrow B} = [I]_{B, B'}$$

Proof. Using Formula (4) of Section 8.4:

$$\begin{aligned} [I]_{B', B} &= [[I(\mathbf{u}_1)]_{B'} \mid [I(\mathbf{u}_2)]_{B'} \mid \cdots \mid [I(\mathbf{u}_n)]_{B'}] \\ &= [[\mathbf{u}_1]_{B'} \mid [\mathbf{u}_2]_{B'} \mid \cdots \mid [\mathbf{u}_n]_{B'}] = P_{B \rightarrow B'} \quad \text{[Formula (3)]} \end{aligned}$$

The proof that $[I]_{B, B'} = P_{B' \rightarrow B}$ is similar. ■

Effect of Changing Bases on Matrices of Linear Operators

Problem

If B and B' are two bases for a finite-dimensional vector space V , and $T : V \rightarrow V$ is a linear operator, what relationship exists between $[T]_B$ and $[T]_{B'}$?

The answer is obtained by writing $T = I \circ T \circ I$ and assigning basis B' to the first and last copies of V , and basis B to the two middle copies, as shown below.

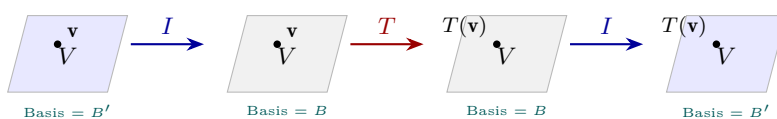


FIGURE 8.5.1

Applying Formula (12) of Section 8.4 to $T = I \circ T \circ I$:

$$[T]_{B'} = [I \circ T \circ I]_{B', B'} = [I]_{B', B} [T]_B [I]_{B, B'} \quad (8)$$

or, with simplified subscript notation,

$$[T]_{B'} = [I]_{B', B} [T]_B [I]_{B, B'} \quad (9)$$

Substituting the transition matrices from Theorem 8.5.1:

$$\boxed{[T]_{B'} = P_{B \rightarrow B'} [T]_B P_{B' \rightarrow B}} \quad (10)$$

Theorem 8.5.2

Let $T : V \rightarrow V$ be a linear operator on a finite-dimensional vector space V , and let B and B' be bases for V . Then

$$[T]_{B'} = P^{-1}[T]_B P \quad (11)$$

where $P = P_{B' \rightarrow B}$ and $P^{-1} = P_{B \rightarrow B'}$.

Warning. In Theorem 8.5.2, $P = P_{B' \rightarrow B}$ (correct) not $P = P_{B \rightarrow B'}$ (incorrect). It helps to observe that in Formula (10) the *exterior* subscripts of the transition matrices match the subscript of the matrix they enclose:

$$[T]_{\underbrace{B'}_{\text{exterior}}} = P_{\underbrace{B \rightarrow B'}_{\text{match}}} [T]_B P_{\underbrace{B' \rightarrow B}_{\text{match}}}$$

Theorem 8.5.3

If V is a finite-dimensional vector space, then two matrices A and B represent the same linear operator (but possibly with respect to different bases) if and only if they are similar. Moreover, if $B = P^{-1}AP$, then P is the transition matrix from the basis used for B to the basis used for A .

EXAMPLE 1 | Similar Matrices Represent the Same Linear Operator

At the beginning of this section, the matrices

$$C = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

represent the same $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ relative to the standard basis B and the basis B' of (2). Verify that C and D are similar by finding P with $D = P^{-1}CP$.

SOLUTION: We need $P = P_{B' \rightarrow B} = [[\mathbf{u}'_1]_B \mid [\mathbf{u}'_2]_B]$. Since B is the standard basis, $[\mathbf{u}'_i]_B = \mathbf{u}'_i$, so

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

Verify:

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & -5 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = D \quad \checkmark$$

Similarity Invariants

Since two matrices are similar iff they represent the same linear operator (Theorem 8.5.3), every similarity invariant of $[T]_B$ is also a similarity invariant of $[T]_{B'}$. In particular, $\det[T]_B = \det[T]_{B'}$, so the **determinant of the linear operator** T is well-defined:

$$\det(T) = \det[T]_B \quad \text{for any basis } B \quad (12)$$

TABLE 1 Similarity Invariants	
Property	Similarity statement
Determinant	$[T]_B$ and $P^{-1}[T]_BP$ have the same determinant.
Invertibility	$[T]_B$ is invertible iff $P^{-1}[T]_BP$ is invertible.
Rank	$[T]_B$ and $P^{-1}[T]_BP$ have the same rank.
Nullity	$[T]_B$ and $P^{-1}[T]_BP$ have the same nullity.
Trace	$[T]_B$ and $P^{-1}[T]_BP$ have the same trace.
Characteristic polynomial	$[T]_B$ and $P^{-1}[T]_BP$ have the same characteristic polynomial.
Eigenvalues	$[T]_B$ and $P^{-1}[T]_BP$ have the same eigenvalues.
Eigenspace dimension	Corresponding eigenspaces have the same dimension.

EXAMPLE 2 | Determinant of a Linear Operator

From the introduction, $[T] = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ and $[T]_{B'} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ represent the same T relative to different bases. As similar matrices they must share the same determinant:

$$\det[T] = \begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix} = 6 \quad \text{and} \quad \det[T]_{B'} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6 \quad \checkmark$$

EXAMPLE 3 | Eigenvalues of a Linear Operator

Find the eigenvalues of $T : P_2 \rightarrow P_2$ defined by

$$T(a + bx + cx^2) = -2c + (a + 2b + c)x + (a + 3c)x^2$$

SOLUTION: Since eigenvalues are similarity invariants, choose any convenient basis. Using $B = \{1, x, x^2\}$, one can verify that

$$[T]_B = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

The eigenvalues of $[T]_B$ are $\lambda = 1$ and $\lambda = 2$ (see Example 7 of Section 5.1), so the eigenvalues of T are $\lambda = 1$ and $\lambda = 2$.

Exercise Set 8.5

In Exercises 1–2, use a property from Table 1 to show that A and B are not similar.

1. a. $A = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$
- b. $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

3. Let $T : R^2 \rightarrow R^2$ be a linear operator, and let B and B' be bases for R^2 for which

$$[T]_B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad P_{B \rightarrow B'} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Find the matrix for T relative to B' .

5. Let $T : R^2 \rightarrow R^2$ be a linear operator, and let B and B' be bases for R^2 for which

$$[T]_{B'} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad P_{B \rightarrow B'} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Find the matrix for T relative to B .

In Exercises 7–14, find $[T]_B$ and use Theorem 8.5.2 to compute $[T]_{B'}$.

7. $T : R^2 \rightarrow R^2$ is defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - 2x_2 \\ -x_2 \end{bmatrix}$, and $B = \{\mathbf{u}_1, \mathbf{u}_2\}$, $B' = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; \quad \mathbf{v}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

9. $T : R^3 \rightarrow R^3$ is defined by $T(x_1, x_2, x_3) = (-2x_1 - x_2, x_1 + x_3, x_2)$, B is the standard basis, and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = (-2, 1, 0), \quad \mathbf{v}_2 = (-1, 0, 1), \quad \mathbf{v}_3 = (0, 1, 0)$$

11. $T : R^2 \rightarrow R^2$ is the rotation about the origin through an angle of 45° , B is the standard basis, and $B' = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \mathbf{v}_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

13. $T : P_1 \rightarrow P_1$ is defined by $T(a_0 + a_1x) = -a_0 + (a_0 + a_1)x$, $B = \{1, x\}$ is the standard basis for P_1 , and $B' = \{\mathbf{q}_1, \mathbf{q}_2\}$, where

$$\mathbf{q}_1 = x + 1, \quad \mathbf{q}_2 = x - 1$$

In Exercises 15–16, find the eigenvalues and bases for the eigenspaces of the linear operator T .

15. $T : P_2 \rightarrow P_2$ is defined by

$$T(a_0 + a_1x + a_2x^2) = (5a_0 + 6a_1 + 2a_2) - (a_1 + 8a_2)x + (a_0 - 2a_2)x^2$$

- a. Find the eigenvalues of T .
- b. Find bases for the eigenspaces of T .

In Exercises 19–21, find the determinant and the eigenvalues of the linear operator T .

19. $T : R^2 \rightarrow R^2$, where $T(x_1, x_2) = (3x_1 - 4x_2, -x_1 + 7x_2)$.
21. $T : P_2 \rightarrow P_2$, where $T(p(x)) = p(x - 1)$.

Working with Proofs

23. Complete the proof below by justifying each step.

Hypothesis: A and B are similar matrices.

Conclusion: A and B have the same characteristic polynomial.

Proof:

$$\begin{aligned}(1) \quad \det(\lambda I - B) &= \det(\lambda I - P^{-1}AP) \\(2) \quad &= \det(\lambda P^{-1}P - P^{-1}AP) \\(3) \quad &= \det(P^{-1}(\lambda I - A)P) \\(4) \quad &= \det(P^{-1}) \det(\lambda I - A) \det(P) \\(5) \quad &= \det(P^{-1}) \det(P) \det(\lambda I - A) \\(6) \quad &= \det(\lambda I - A)\end{aligned}$$

In Exercises 25–28, prove that the stated property is a similarity invariant.

25. Trace

27. Nullity